

MadePlus Smart Shoe - Walking Session Report

1. Session Overview - Total distance: 239 m - Duration: 147 s ($\approx$ 2 min 27 s) - Average pace (including pauses): 10 min 15 s per km - Total steps: 59 - Average stride length written in CSV: 4.052 m (this is distance/steps, a session-average value rather than per-step).

2. Cadence Behaviour

- 0-10 s: Cadence starts at 0, then rises into the 20-30 range as you begin walking (warm up phase). - 10-60 s: Cadence stabilises mostly between 30-45, corresponding to a comfortable, steady walk. - 60-95 s: Cadence dips into the 20-30 band with a brief low/near-zero patch around 90 s, suggesting a short slow-down or micro-pause. - 112-121 s: Clear fast segment. Cadence jumps into roughly the 55-75 range with a maximum near 80, this is the deliberate "walk faster" burst. - 121-147 s: Cadence drops again with a few slower steps and then almost no movement as you cool down and stop.

3. Step Pattern

- The cumulative step curve climbs cleanly from 0 to 59 with no downward jumps, showing that step detection is stable (no double count or reset). - Steeper sections of the curve coincide with the high-cadence burst near 110-120 s. - Flatter sections appear around the slower/micro-pause zone (~90 s) and near the end, matching what is seen in the cadence graph.

4. Foot Motion (Pitch)

- The pitch signal oscillates between roughly 15° and 60° during steady walking, representing a normal heel-to-toe step cycle. - In the slower middle phase the oscillations are slightly smaller and less frequent, consistent with shorter, slower steps. - During the fast burst the oscillations become more frequent and sometimes larger indicating a more dynamic, brisk gait. - Near the end the pattern becomes irregular and then settles as you slow down and stop.

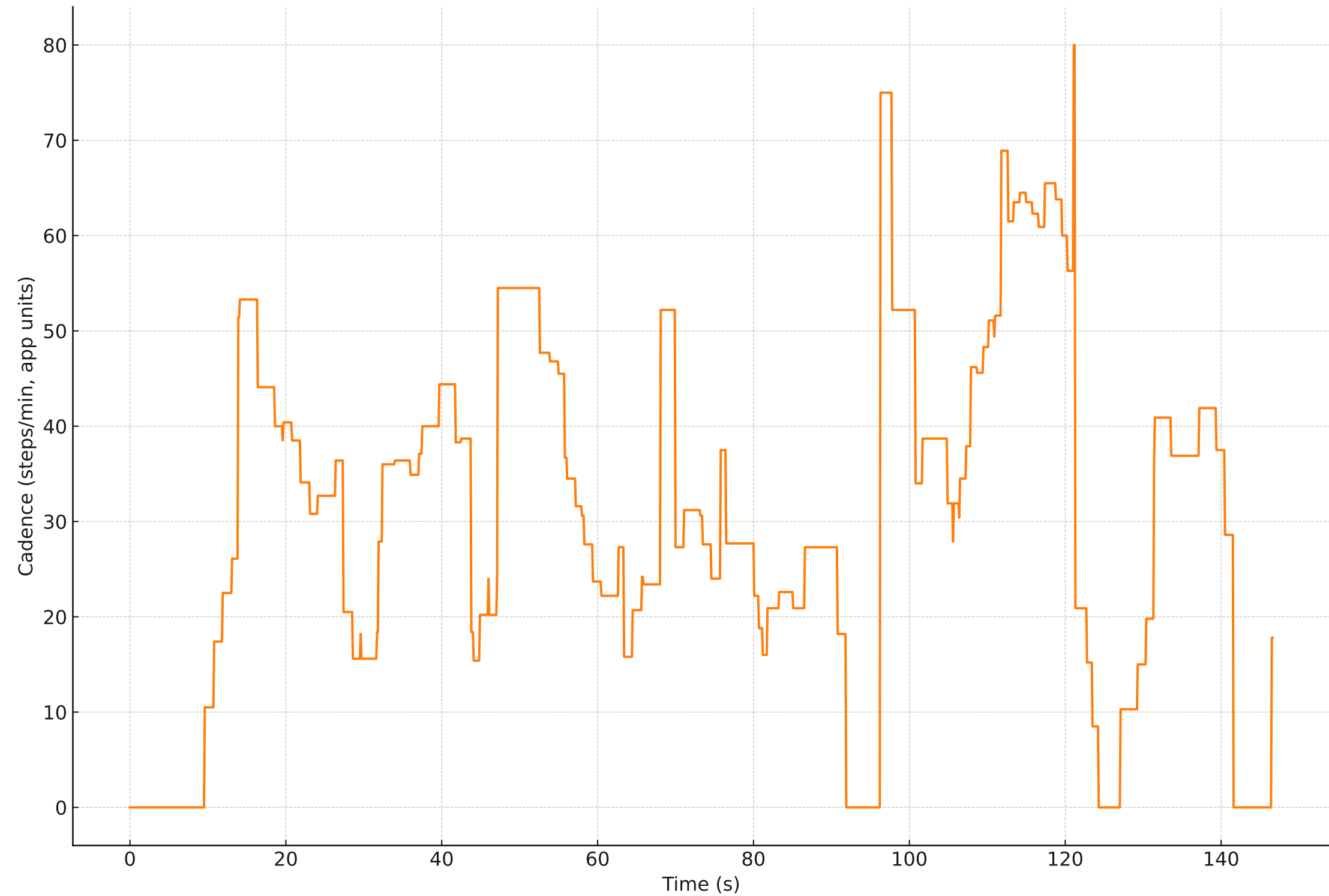
5. Temperature & Sensor Stability

- Temperature drifts smoothly from about 28.0–28.1 °C at the start down to about 25.4–25.5 °C by the end of the session. - There are no sudden jumps, suggesting the temperature sensor is stable and the change is due to normal environmental effects or airflow while walking.

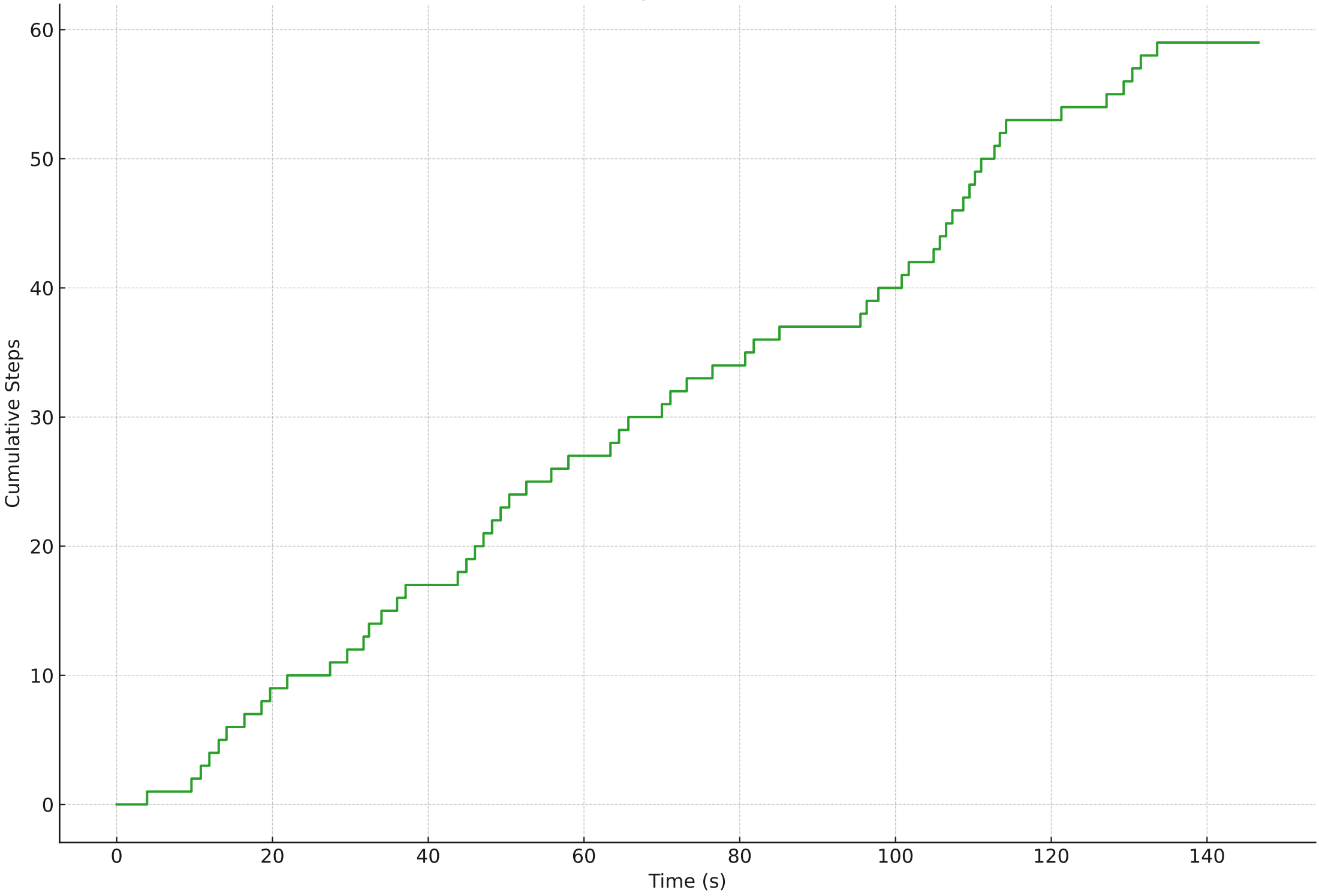
6. Overall Interpretation

- You begin from rest, start walking and quickly reach a comfortable moderate pace. - After roughly a minute you ease off a bit and there is a short micro-pause or turning phase, reflected in cadence and step timing. - You then perform one clearly faster segment near 112–121 s, with high cadence and tightly spaced steps - this is the deliberate faster block. - Finally you reduce speed and come to a stop, giving the pattern: casual → slightly slower with a brief pause → brisk burst → cool-down and stop. - Step detection is clean, and the strideM field in this CSV represents a single session-average stride length rather than a per-step value.

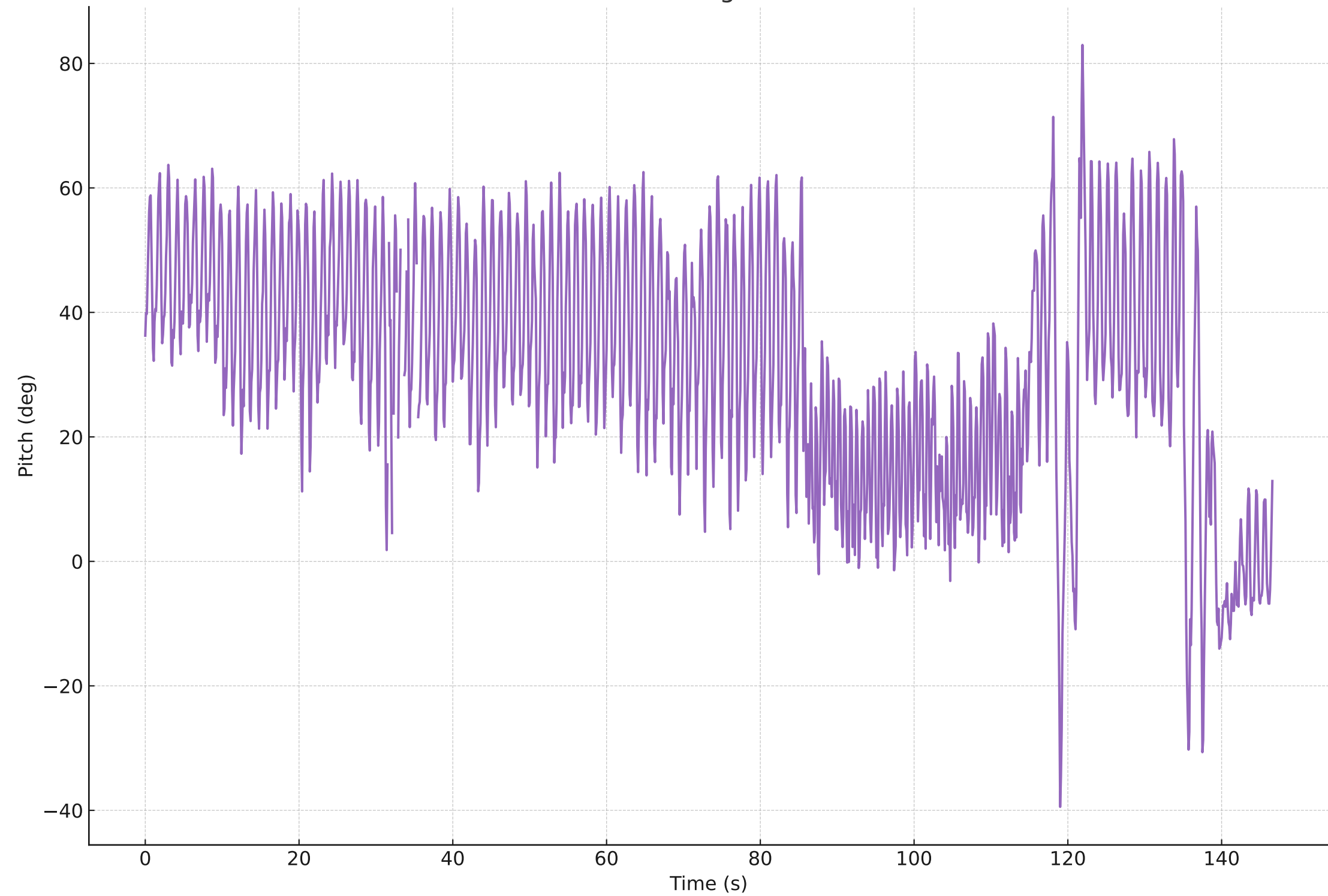
Cadence vs Time



Steps vs Time



Foot Pitch Angle vs Time



Temperature vs Time

